

SOFTWARE ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection

Telemedicine Slot Booking & Prescription Portal

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Architecture Selection – Layered Architecture for the Telemedicine Slot Booking & Prescription Portal.

The system is divided into three main layers:

- 1) Presentation Layer
- 2) Business Layer and
- 3) Data Layer.

- The Presentation Layer contains the Patient Portal UI and Physician Dashboard.
- The Business Layer contains the Booking Manager, Teleconsult Service, and Prescription Service.
- The Data Layer contains the Slot & Booking Database, Encryption Module, and Prescription Store.

Reason 1 – Separation of Concerns

Layered architecture separates user interaction, business processing, and data management. This makes the telemedicine system easier to understand, maintain, and modify. For example, changes to the patient or physician interface can be made in the Presentation Layer without directly modifying the booking or prescription logic.

Reason 2 – Suitable for Core Telemedicine Functions

The architecture clearly separates important functions such as slot booking, video consultation, and prescription management. The Booking Manager can handle appointment-related operations, while the Teleconsult Service manages consultation functionality and the Prescription Service manages digital prescriptions. This organization supports the requirements identified in Lab 1.

Security Advantage

The separation of layers allows security mechanisms to be centralized in the appropriate components, such as the Encryption Module and Prescription Store. Sensitive consultation and prescription information can therefore be protected before being stored or exchanged between components.

Performance Benefit

The layered structure allows frequently accessed operations such as viewing available slots and booking consultations to be handled by dedicated business and data components. This can reduce unnecessary processing and help the system respond efficiently to booking requests.

UML DIAGRAM

